Who Pays the Bill?

Consolidated Research Checkpoint v8:
Exact 19/9 Infinite-Viability Lower Bound and the Multiscale Zero-Repair Frontier

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22 August 2026

Abstract

This paper consolidates the latest mathematical status of the causal information-bill programme. For equal-entropy finite sources with unequal varentropy, the sharp second-order causal residual bill is

$$E_n(P,Q) = \sqrt{\frac{2}{\pi}}\,|\sigma_P - \sigma_Q|\sqrt{n} + o(\sqrt{n}).$$

For the higher-order dyadic parity ladder, if the first unmatched information moment is of order r, an entirely atomic B-spline construction gives the rigorous causal upper bound

$$E_n(P_R, Q_R) = O(n^{1/r}).$$

A matching lower bound is rigorous for stationary independent reservoirs, but that architecture is not universal.

The first nontrivial higher-order case, $r = 3$, reduces to an exact history-indexed zero-repair viability problem. Earlier exact LPs through depth eleven gave $\mathcal{B}_{11} = 1.897576589\ldots < 2$, and explicit mean-2 root laws survived to that finite depth. Those facts are now known to be finite-horizon phenomena rather than evidence for an infinite mean-2 solution.

The decisive new result of this version is an exact, support-independent infinite-viability lower certificate. A closed low-tail envelope first yields

$$\mathbb{E}J \geq \frac{17}{9},$$

one more generation of coupled viability inequalities yields

$$\mathbb{E}J \geq \frac{55}{27},$$

and an exact rational two-generation LP-dual certificate gives

$$\boxed{\mathbb{E}J \geq \frac{19}{9} > 2}$$

for the root of every infinite zero-repair history-indexed tree. Consequently

$$\boxed{\mathcal{B}_\infty \geq \frac{19}{9} > 2},$$

so the previous candidate $\mathcal{B}_\infty = 2$, every infinite mean-2 root, and the proposed universal affine lift forcing $\mathcal{B}_n \leq 2$ are all ruled out.

The structural no-go results from v7 remain in force: no finite exact zero-repair automaton exists for any finite number of modes; no scalar state-only Bellman potential carries a strictly positive uniform drift; no depth-only or score-only finite-mean compression solves the infinite problem; and neither one prefix-majorization Hall envelope nor any finite family of finite-mean Hall envelopes closes. Exact Hall transfer, transform martingale, cubic transform gap, and third-moment flux identities remain available.

The live dichotomy is therefore strictly sharper:

$$\boxed{\frac{19}{9} \le \mathcal{B}_\infty < \infty} \qquad \text{versus} \qquad \boxed{\mathcal{B}_n \to \infty}.$$

No completed proof selecting between these two alternatives is claimed here. The active research frontier is to push the closed-tail/dual hierarchy beyond 19/9, or construct a genuinely order-sensitive infinite multiscale repair with finite root mean at some value at least 19/9.

Status convention. Statements labelled as theorems, propositions, lemmas, or corollaries are analytic results. Numerical LP data are evidence only. Passages marked **ACTIVE RESEARCH FRONTIER** are intentionally open and isolate the exact places where the current proof attempt is still being attacked.

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1 The causal bill

Let P, Q be finite probability distributions. Their single-symbol surprisals are

$$\imath_P(x) := -\log_2 P(x), \qquad \imath_Q(y) := -\log_2 Q(y).$$

Write

$$H(P) = \mathbb{E}\imath_P(X), \qquad V(P) = \mathrm{Var}(\imath_P(X)),$$

and analogously for Q.

We work with the history-indexed causal-tree contract C3-C.

Definition 1.1 (Exact causal action tree)**.** *A random action tree of depth n is exact for (P, Q) if, for every deterministic action word $a_1 \cdots a_n \in \{P, Q\}^n$, the labels observed along the corresponding path have the product law prescribed by the action word.*

Let

$$C_n^{\mathrm{causal}}(P, Q)$$

be the minimum Shannon entropy of a latent random seed generating such an exact depth-n tree.

In the equal-entropy case

$$H(P) = H(Q) = h,$$

define the residual causal bill

$$\boxed{E_n(P, Q) := C_n^{\mathrm{causal}}(P, Q) - nh.}$$

The central question is not whether exact simulation is possible; it is how much *additional* latent information is forced by exact causality.

2 Established structural facts

The following statements are part of the established framework.

2.1 Prefix lower bound and subadditivity

For all m, n,

$$C_{m+n}^{\mathrm{causal}} \le C_m^{\mathrm{causal}} + C_n^{\mathrm{causal}}.$$

Hence

$$C_\infty^{\mathrm{causal}} := \lim_{n\to\infty} \frac{C_n^{\mathrm{causal}}}{n}$$

exists by Fekete's lemma.

In the equal-entropy case,

$$E_n \ge 0,$$

and E_n is subadditive.

2.2 Projective closure

There exists a single infinite, generally nonstationary exact causal tree whose finite prefixes attain the asymptotic causal rate:

$$\inf_T \limsup_{n\to\infty} \frac{H(T_{\le n})}{n} = C_\infty^{\mathrm{causal}}.$$

A block-diagonal construction with increasing near-optimal finite blocks gives an infinite projectively consistent substrate.

The stronger stationary or shift-invariant version is a separate problem.

2.3 Stream-to-tree anti-diagonal lift

Suppose a joint law of W^n, X^n satisfies

$$W^n \sim Q^n, \qquad X^n \sim P^n,$$

and the one-way causality condition

$$\mathrm{Law}(X_{1:k} \mid W_{1:n}) = \mathrm{Law}(X_{1:k} \mid W_{1:k}) \qquad (k \leq n).$$

Then the anti-diagonal assignment of P-requests to the X-stream and Q-requests to the reversed W-stream yields an exact causal action tree. Consequently

$$C_n^{\text{causal}}(P, Q) \leq H(X^n, W^n),$$

and in the equal-entropy case

$$\boxed{E_n(P, Q) \leq H(X^n \mid W^n).}$$

Thus exact causal stream couplings are constructive upper certificates for the tree problem.

3 Sharp second-order theorem

Assume

$$H(P) = H(Q) = h$$

and let

$$\sigma_P^2 = V(P), \qquad \sigma_Q^2 = V(Q).$$

The sharp second-order theorem is

$$\boxed{E_n(P, Q) = \sqrt{\frac{2}{\pi}}\, |\sigma_P - \sigma_Q|\, \sqrt{n} + o(\sqrt{n}).}$$

Two mechanisms meet exactly.

3.1 Adaptive lower bound

For a deterministic adaptive policy π, the transcript is a deterministic function of the shared seed. Atomwise,

$$-\log_2 \mathbb{P}(S = s) \geq I_\pi(Y^\pi(s)).$$

Hence

$$H(S) \geq \mathbb{E} \max_\pi I_\pi(Y^\pi).$$

Threshold policies asymptotically generate an oscillating Brownian motion. The optimal split-normal envelope has mean

$$\sqrt{\frac{2}{\pi}}\, |\sigma_P - \sigma_Q|,$$

giving

$$\liminf_{n \to \infty} \frac{E_n}{\sqrt{n}} \geq \sqrt{\frac{2}{\pi}}\, |\sigma_P - \sigma_Q|.$$

3.2 Sorted-arithmetic upper bound

The upper construction sorts length-L source atoms by surprisal, reuses a uniform arithmetic phase, and produces exact monotone coupling of successive block information spectra.

After controlling the rare boundary defect and finite-bit exactization, the resulting block increment Z_L satisfies

$$\frac{\mathbb{E}Z_L^2}{L} \longrightarrow (\sigma_P - \sigma_Q)^2.$$

A triangular-array functional CLT and uniform integrability of the running maximum then give

$$\limsup_{n\to\infty} \frac{E_n}{\sqrt{n}} \le \sqrt{\frac{2}{\pi}}\,|\sigma_P - \sigma_Q|.$$

4 The crash pair

A convenient equal-entropy unequal-varentropy test pair is

$$P = \left(\frac{1}{2}, \frac{1}{4}, \frac{1}{4}\right),$$

and

$$Q = (q, q, 1-2q),$$

where q is the unique root giving

$$H(Q) = \frac{3}{2}.$$

Numerically,

$$q = 0.41003242818198557608\ldots,$$

$$1 - 2q = 0.17993514363602884784\ldots.$$

Then

$$H(P) = H(Q) = \frac{3}{2},$$

$$\sigma_P = \frac{1}{2}, \qquad \sigma_Q = 0.456450775263963\ldots,$$

and therefore

$$\boxed{E_n = 0.0347472540518178 64\ldots\sqrt{n} + o(\sqrt{n}).}$$

This pair is the main second-order stress test.

5 Why the higher-order problem is different

Suppose now

$$H(P) = H(Q), \qquad V(P) = V(Q).$$

The sharp second-order coefficient vanishes:

$$\sqrt{\frac{2}{\pi}}|\sigma_P - \sigma_Q| = 0.$$

A first Edgeworth heuristic suggests that if the first unmatched centered information cumulant is order r, then normalized block spectra differ at scale

$$L^{-(r-2)/2},$$

so the unnormalized monotone block mismatch is expected at scale

$$L^{(3-r)/2}.$$

A delayed single-scale block construction would then balance

$$L \quad \text{against} \quad \sqrt{n}\, L^{(2-r)/2},$$

producing

$$L \asymp n^{1/r}$$

and therefore the candidate residual scale

$$n^{1/r}.$$

This heuristic is useful but *not* the basis of the rigorous higher-order upper bound below. In particular, lattice-phase corrections can dominate naive Cornish–Fisher constants.

6 The dyadic parity ladder

Fix $R \geq 1$, and put

$$r := R + 1.$$

Define the surprisal laws

$$\mu_{P_R} = 2^{-R} \sum_{\substack{0 \leq j \leq R+1 \\ j \text{ even}}} \binom{R+1}{j} \delta_{R+j},$$

and

$$\mu_{Q_R} = 2^{-R} \sum_{\substack{0 \leq j \leq R+1 \\ j \text{ odd}}} \binom{R+1}{j} \delta_{R+j}.$$

They are realized by finite atomic probability vectors because a surprisal level $R+j$ corresponds to atom probability $2^{-(R+j)}$, and the multiplicities are integers.

Their generating functions satisfy

$$G_{P_R}(z) - G_{Q_R}(z) = 2^{-R} z^R (1-z)^{R+1}.$$

Equivalently, up to translation,

$$\boxed{\mu_{P_R} - \mu_{Q_R} = 2^{-(r-1)} \Delta^r.}$$

Proposition 6.1 (Moment matching)**.** *The first $r-1$ information moments agree, and the first unmatched moment is order r.*

Proof. The difference of generating functions contains the factor $(1-z)^r$, so the first $r-1$ derivatives agree at $z = 1$, while the rth derivative does not. □

6.1 The first nontrivial higher-order case

For $R = 2$, hence $r = 3$,

$$\boxed{\mu_P = \frac{1}{4}\delta_2 + \frac{3}{4}\delta_4, \qquad \mu_Q = \frac{3}{4}\delta_3 + \frac{1}{4}\delta_5.}$$

Their common information mean is

$$h = \frac{7}{2},$$

and their common varentropy is

$$V = \frac{3}{4}.$$

The centered third cumulants are

$$\kappa_3(P) = -\frac{3}{4}, \qquad \kappa_3(Q) = +\frac{3}{4}.$$

Thus this is an exact

$$H_P = H_Q, \qquad V_P = V_Q, \qquad \kappa_3(P) \neq \kappa_3(Q)$$

crash pair.

7 Lattice phase is real

For $R = 2$, the centered block information laws are explicit:

$$A_L = -\frac{3L}{2} + 2K, \qquad K \sim \mathrm{Bin}(L, 3/4),$$

and

$$B_L = -\frac{L}{2} + 2J, \qquad J \sim \mathrm{Bin}(L, 1/4).$$

Exact monotone transport shows two distinct subsequential W_2-limits for even and odd L. Numerically,

$$W_2(A_L, B_L) \to 0.816\ldots$$

along even L, while

$$W_2(A_L, B_L) \to 1.054\ldots$$

along odd L.

Hence a universal third-order constant cannot be read directly from a nonlattice Cornish–Fisher formula. The exponent may survive, but the constant is arithmetic.

8 Atomic B-spline buffer

Let

$$U_1, \ldots, U_r \overset{\text{iid}}{\sim} \mathrm{Unif}\{0, \ldots, m-1\}, \qquad T := U_1 + \cdots + U_r.$$

Write

$$\mathbb{P}(T = t) = \frac{c_t}{m^r}.$$

At macrostate t, create

$$c_t 2^t$$

microatoms, each of probability

$$\frac{1}{m^r 2^t}.$$

Thus every buffer microatom at level t has surprisal

$$\boxed{r \log_2 m + t.}$$

The macro-coordinate t is therefore a true information coordinate.

Proposition 8.1 (Exact buffer entropy)**.** *The buffer entropy is*

$$\boxed{H(B_m) = r \log_2 m + \frac{r}{2}(m-1).}$$

Proof. The microatom surprisal is $r \log_2 m + T$, and

$$\mathbb{E}T = r\,\frac{m-1}{2}.$$

$\square$

The box generating function is

$$U_m(z) = \frac{1-z^m}{m(1-z)}.$$

Therefore

$$(G_{P_R} - G_{Q_R})U_m(z)^r = 2^{-(r-1)} m^{-r} z^{r-1}(1-z^m)^r.$$

Theorem 8.2 (Exact atomic residual mass)**.** *For the extended atomic laws*

$$P_R \times B_m \qquad \textit{and} \qquad B_m \times Q_R,$$

the common equal-surprisal atoms can be paired deterministically, and the unmatched mass on each side is exactly

$$\boxed{\varepsilon_m = m^{-r}.}$$

Proof. The signed residual coefficients are

$$2^{-(r-1)} m^{-r} (-1)^k \binom{r}{k}.$$

The total variation of the signed residual is

$$2m^{-r},$$

so the positive and negative residual masses are each m^{-r}. At a fixed surprisal level, all atomic probabilities are identical; hence the common part admits an exact atom-by-atom bijection. $\square$

Conditioned on the residual output,

$$\boxed{H_{\text{res}} = (r-1) + \frac{r}{2} m.}$$

9 Constructive causal upper hierarchy

Let

$$B_0 \sim B_m$$

independently of the Q_R-stream.

At each time t, use the exact atomic coupling

$$(B_{t-1}, W_t) \longmapsto (X_t, B_t),$$

deterministically on the matched mass and with fresh randomness only on the residual event.

The transition sends

$$B_m \times Q_R$$

exactly to

$$P_R \times B_m.$$

Hence

$$(X_1, \dots, X_n, B_n) \sim P_R^n \times B_m,$$

and the construction is one-way causal.

By the chain rule,

$$H(X^n \mid W^n) \leq H(B_m) + n\,\varepsilon_m H_{\mathrm{res}}.$$

Theorem 9.1 (Explicit parity-ladder upper bound)**.** *For every integer $m \geq 1$,*

$$\boxed{E_n(P_R, Q_R) \leq r \log_2 m + \frac{r}{2}(m-1) + \frac{n}{m^r}\left(r - 1 + \frac{r}{2}m\right).}$$

Therefore

$$\boxed{E_n(P_R, Q_R) = O(n^{1/r}).}$$

Moreover,

$$\boxed{\limsup_{n\to\infty} \frac{E_n(P_R, Q_R)}{n^{1/r}} \leq \frac{r^2}{2}(r-1)^{-(r-1)/r}.}$$

Proof. The displayed finite-n bound follows from the exact initial buffer entropy and the exact residual transition entropy. The leading terms are

$$\frac{r}{2}m + \frac{r}{2}\frac{n}{m^{r-1}}.$$

Choosing

$$m \sim ((r-1)n)^{1/r}$$

gives the stated asymptotic constant. $\square$

For $r = 3$,

$$\boxed{\limsup_{n\to\infty} \frac{E_n}{n^{1/3}} \leq \frac{9}{2}2^{-2/3} = 2.834822362\ldots.}$$

This upper bound is independent of the delayed-block startup mechanism.

10 REAL kernels and the actual smoothing hierarchy

The REAL family contains compactly supported kernels

$$\rho_k(x) = \frac{k+1}{2}(1-|x|)_+^k.$$

For $r = 2$, the optimal discrete buffer profile is a box, the discrete analogue of ρ_0.

For $r = 3$, the exact optimizer of the reduced W_1-buffer problem is the discrete triangular law

$$\pi_M(s) = \frac{M+1-|s|}{(M+1)^2}, \qquad |s| \le M,$$

whose continuum profile is exactly

$$\rho_1(x) = (1-|x|)_+.$$

Thus

$$\boxed{\text{REAL } \rho_1 = \text{the exact third-order information smoother.}}$$

However, for $r \ge 4$, the family $(1-|x|)_+^k$ is not the correct higher-order hierarchy. The cusp at the origin destroys one order of finite-difference smoothing. The correct natural family is given by cardinal B-splines, that is, repeated convolutions of a box.

Hence the REAL connection is exact at the first two levels but does not, without modification, provide the full r-hierarchy.

11 Stationary independent-buffer lower bound

Suppose a stationary architecture has an independent information reservoir B, with surprisal

$$J_B = -\log_2 \mathbb{P}(B)$$

and

$$H(B) = \mathbb{E}J_B.$$

Then

$$|\varphi_B(t)| = |\mathbb{E}e^{itJ_B}| \ge 1 - |t|H(B).$$

For the parity ladder,

$$\boxed{|\varphi_{P_R}(t) - \varphi_{Q_R}(t)| = 2|\sin(t/2)|^r.}$$

For the extended input and output information profiles

$$A = B \times Q_R, \qquad Y = P_R \times B,$$

we have

$$|\varphi_Y(t) - \varphi_A(t)| = |\varphi_B(t)|\,|\varphi_{P_R}(t) - \varphi_{Q_R}(t)|.$$

Kantorovich–Rubinstein yields

$$W_1(\mu_A, \mu_Y) \ge \frac{|\varphi_Y(t) - \varphi_A(t)|}{|t|}.$$

Choosing

$$t \asymp \frac{1}{H(B)}$$

gives

$$W_1(\mu_A, \mu_Y) \gtrsim H(B)^{-(r-1)}.$$

Using the equal-entropy information-profile converse

$$H(Y \mid A) \geq \frac{1}{2} W_1(\mu_A, \mu_Y),$$

we obtain

$$\text{cost} \geq H(B) + c_r \frac{n}{H(B)^{r-1}}.$$

Therefore, within the stationary independent-buffer architecture,

$$\boxed{\text{cost} = \Omega(n^{1/r}).}$$

Combined with the B-spline upper,

$$\boxed{\text{stationary independent-buffer class: } \Theta(n^{1/r}).}$$

This is a genuine theorem, but it is not yet a theorem for the full C3-C class.

12 Why the stationary lower is too narrow

For $R = 2$, allowing the next buffer to correlate with the current output produces a formal stationary profile

$$\boxed{B_\infty(z) = \frac{z^2(z+1)}{3-z}.}$$

Its coefficients are

$$\beta_2 = \frac{1}{3}, \qquad \beta_j = \frac{4}{3^{j-1}}, \quad j \geq 3.$$

At the one-step information-profile level it achieves zero repair with finite mean.

Thus the universal statement

"every causal buffer must itself have size $n^{1/3}$"

is false.

The problem is that the child buffer law depends on the output branch, and therefore the one-step stationary profile cannot simply be iterated while preserving the complete iid output process.

13 The zero-repair history-indexed tree

For each output history

$$h \in \{0,1\}^{\leq n},$$

let B_h be a probability law on $\mathbb{Z}_{\geq 0}$.

For $R = 2$, define the exact zero-repair recursion

$$\boxed{\mu_Q * B_h = \frac{1}{4}\delta_2 * B_{h0} + \frac{3}{4}\delta_4 * B_{h1}.} \tag{Z}$$

Definition 13.1 (Finite-horizon zero-repair bill)**.** *Define*

$$\boxed{\mathcal{B}_n := \inf \left\{ \mathbb{E}_{B_\varnothing} J : \{B_h\}_{|h| \leq n} \textit{satisfies } (\text{Z}) \right\}.}$$

This object isolates the profile-level cost of exact history-compatible information conservation.

14 Exact finite LP data

The recursion (Z) is linear in the buffer laws, so $\mathcal{B}_n$ can be computed exactly as a finite LP after imposing a sufficiently large support cutoff.

The first values are:

n	$\mathcal{B}_n$	$\mathcal{B}_n/n^{1/3}$
1	0.666666667	0.6667
2	1.000000000	0.7937
3	1.185185185	0.8218
4	1.296296296	0.8165
5	1.432098765	0.8375
6	1.506172840	0.8289
7	1.596707819	0.8347
8	1.695473251	0.8477
9	1.772290809	0.8520
10	1.846364883	0.8570
11	1.897576589	0.8532

The exact-looking rational values include

$$\mathcal{B}_{10} = \frac{1346}{729}, \qquad \mathcal{B}_{11} = \frac{4150}{2187}.$$

The root optimum is stable when the artificial support cutoff is enlarged, so these values are not a simple truncation artifact.

15 The finite-horizon mirage near two

The exact depth-11 data once suggested a possible finite limit near 2. The final gaps were

n	$2 - \mathcal{B}_n$
8	0.304526749
9	0.227709191
10	0.153635117
11	0.102423411

and

$$\frac{2 - \mathcal{B}_{11}}{2 - \mathcal{B}_{10}} = \frac{2}{3}.$$

These data were always insufficient to justify an asymptotic conclusion. The exact infinite-viability certificate proved below now shows that the apparent attraction to 2 cannot persist:

$$\boxed{\mathcal{B}_\infty \geq \frac{19}{9} > 2.}$$

Thus any finite-depth approach toward 2 must eventually reverse or cross above 2; the depth-11 values are pre-asymptotic.

The live alternatives are now

$$\boxed{H_1 : \quad \frac{19}{9} \leq \mathcal{B}_\infty < \infty,}$$

and

$$\boxed{H_2 : \quad \mathcal{B}_n \to \infty.}$$

16 Mean-2 roots survive finite depth but are globally impossible

Several explicit root laws of mean exactly 2 remain feasible for the zero-repair LP through depth at least 11, including

$$B_\varnothing(z) = \left(\frac{1+z}{2}\right)^4$$

and

$$B_\varnothing(z) = \left(\frac{1+z+z^2}{3}\right)^2.$$

This is still useful as a stress test for finite-horizon algorithms. It is no longer a live infinite-tree hypothesis. The exact lower theorem below gives

$$\mathbb{E}J_\varnothing \geq \frac{19}{9} > 2$$

for every infinite viable tree. Hence no mean-2 law belongs to $\mathcal{K}_\infty$.

17 Greedy allocation and why it fails

Given a parent law $b_j = \mathbb{P}(B_h = j)$, define the total input information profile

$$r_u = \frac{3}{4}b_{u-3} + \frac{1}{4}b_{u-5}.$$

A child decomposition is determined by choosing a submeasure

$$0 \leq c_u \leq r_u, \qquad \sum_u c_u = \frac{1}{4},$$

for the low-P branch.

Then

$$b_j^{(0)} = 4c_{j+2},$$

and

$$b_j^{(1)} = \frac{4}{3}\left(r_{j+4} - c_{j+4}\right).$$

The natural greedy rule chooses the lowest information levels until quarter mass is collected. This works for many levels but eventually creates an infeasible child.

For one natural geometric profile, the first failure appears at history

$$01101101,$$

where the child law becomes

$$\frac{2}{3}\delta_0 + \frac{1}{3}\delta_2.$$

Then the next input puts mass

$$\frac{3}{4}\cdot\frac{2}{3} = \frac{1}{2}$$

at information level 3, more than the low-output branch can absorb without requiring negative information in the high branch.

Thus

$$\boxed{\text{greedy low-level allocation is not an infinite solution.}}$$

The infinite problem requires anticipative viability, not myopic matching.

18 No finite-state stationary zero-repair transducer

Consider an exact hidden-state transducer

$$J_t = J_{t-1} + I_Q(W_t) - I_P(X_t),$$

with

$$W_t \mid J_{t-1} \sim Q, \qquad X_t \mid J_{t-1} \sim P.$$

Because

$$H(P) = H(Q),$$

we have

$$\mathbb{E}[J_t \mid J_{t-1}] = J_{t-1}.$$

Hence J_t is a martingale.

Lemma 18.1 (No stationary integrable zero-repair state)**.** *There is no nontrivial stationary integrable state law for such a zero-repair transducer.*

Proof. Assume

$$J_t \overset{d}{=} J_{t-1}$$

and

$$\mathbb{E}J_t < \infty.$$

Take the strictly convex linearly growing function

$$\psi(x) = \sqrt{1 + x^2}.$$

Conditional Jensen gives

$$\mathbb{E}[\psi(J_t) \mid J_{t-1}] \geq \psi(J_{t-1}).$$

Stationarity forces equality after expectation, hence equality holds in strict Jensen:

$$J_t = J_{t-1} \quad \text{a.s.}$$

For the $R = 2$ pair,

$$I_Q - I_P \in \{-1, +1, +3\},$$

so zero increment is impossible. Contradiction. □

Exhaustive finite-state LP searches with up to forty state levels also find no stationary exact zero-repair transducer.

Thus any H_1 construction must be genuinely history-dependent and multiscale.

19 Complete finite-mode obstruction

The finite-mode obstruction can be proved for every finite number of modes; the earlier $K \leq 3$ computation is not needed.

Theorem 19.1 (No finite exact zero-repair mode automaton)**.** *There do not exist a finite integer $K \geq 1$, probability laws $B^{(1)}, \ldots, B^{(K)}$ on $\mathbb{Z}_{\geq 0}$, and deterministic transition maps*

$$\tau_0, \tau_1 : \{1, \ldots, K\} \to \{1, \ldots, K\}$$

such that for every mode s

$$\mu_Q * B^{(s)} = \frac{1}{4}\delta_2 * B^{(\tau_0(s))} + \frac{3}{4}\delta_4 * B^{(\tau_1(s))}.$$

Proof. Let $G_s(z)$ be the probability generating function of $B^{(s)}$. Multiplying the zero-repair identity by 4 and removing the common factor z^2 gives

$$z(3+z^2)G_s(z) = G_{\tau_0(s)}(z) + 3z^2 G_{\tau_1(s)}(z).$$

Let $G = (G_1, \dots, G_K)^\top$, and let L, H be the row-selector matrices corresponding to τ_0, τ_1. Then

$$M(z)G(z) = 0, \qquad M(z) := z(3+z^2)I - L - 3z^2 H.$$

The determinant $\det M(z)$ is a polynomial. Its unique contribution of degree $3K$ comes from choosing the z^3 term on every diagonal entry. Hence

$$\det M(z) = z^{3K} + \text{lower-degree terms},$$

so $\det M$ is not the zero polynomial. Therefore $M(z)$ is invertible for all but finitely many $z \in (0,1)$. At each such z,

$$G(z) = 0.$$

This is impossible because every probability PGF satisfies $G_s(z) > 0$ for $0 < z < 1$. Thus no finite exact mode automaton exists. □

Remark 19.2. *The same degree argument applies to finite randomized mode transitions as long as the child-profile closure remains linear with fixed finite transition matrices. Thus the obstruction is finite-dimensional, not merely deterministic-combinatorial.*

PROVED. Finite-memory zero-repair recycling is ruled out for every finite number of exact profile modes. Any H_1 construction must use an actually infinite hierarchy of scales or histories.

20 Harvey–Peres self-similarity

For $R = 2$,

$$G_P(z)^2 - G_Q(z)^2 = \frac{1}{4}\left(G_P(z^2) - G_Q(z^2)\right).$$

At the two-symbol information-spectrum level,

$$\mu_P^{*2} = \frac{1}{16}\delta_4 + \frac{6}{16}\delta_6 + \frac{9}{16}\delta_8,$$

and

$$\mu_Q^{*2} = \frac{9}{16}\delta_6 + \frac{6}{16}\delta_8 + \frac{1}{16}\delta_{10}.$$

The ordered common mass is $3/4$, and the unmatched mass is $1/4$. After conditioning and dividing the information coordinate by 2, the residual profile is again the original parity-ladder pair.

Harvey–Peres ordered matching turns this signed self-similarity into a positive residual decomposition. Therefore:

$$\boxed{\text{static positivity is not the open problem.}}$$

The remaining difficulty is unilateral/prefix compatibility of the future subtrees attached to equal-total-surprisal words such as 01 and 10.

21 Why simple Bellman reductions fail

Several plausible lower-bound reductions have been tested.

21.1 Terminal or prefix stochastic dominance

Requiring only

$$B + I_Q^{(t)} \succeq_{\mathrm{st}} I_P^{(t)}$$

for all prefixes gives a bounded cost near 0.894, far below the full history-indexed zero-repair LP.

Thus the obstruction is not terminal inventory.

21.2 Depth-only dual

A Bellman dual depending only on depth and buffer information level is trivial.

21.3 Count-only history compression

Forcing the buffer to depend only on depth and the number of high-P outputs changes the primal optimum drastically. The ordering of the history matters.

21.4 Fixed finite-memory dual

Dual certificates depending on only the final ℓ output symbols saturate for each fixed ℓ.

Therefore a sharp lower certificate, if one exists, must use memory across growing scales.

21.5 No canonical smooth cubic dual

Raw HiGHS dual multipliers are strongly degenerate. They do not collapse to a canonical smooth function of

$$j/n^{1/3}.$$

The optimal child allocation itself develops disconnected "islands" at multiple information scales, which is much more consistent with a hierarchical self-similar certificate than with a single cubic barrier.

22 A complete no-go theorem for scalar Bellman drift

A natural dual attempt assigns a potential $\phi(j)$ to the current buffer state and asks for a positive uniform drift. Periodicity is not the reason this fails; the entire state-only ansatz fails.

Theorem 22.1 (No positive scalar state-only Bellman drift)**.** *Let $\phi : \mathbb{Z}_{\geq 0} \to \mathbb{R}$. Suppose there are constants c, λ_x for $x \in \{2, 4\}$, and μ_w for $w \in \{3, 5\}$ such that, whenever $j + w - x \geq 0$,*

$$\phi(j + w - x) - \phi(j) \geq c + \lambda_x + \mu_w, \tag{B}$$

and suppose the gauge is normalized by

$$\sum_x P(x)\lambda_x = 0, \qquad \sum_w Q(w)\mu_w = 0.$$

Then necessarily $c \leq 0$.

Proof. Set $d_j := \phi(j + 1) - \phi(j)$. Among the admissible increments $w - x$ are $+1$ and -1. The corresponding instances of (B) give a uniform lower bound and a uniform upper bound on d_j for all sufficiently large j. Hence the Cesàro averages

$$a_M := \frac{1}{M} \sum_{j=0}^{M-1} d_j$$

are bounded. Along a subsequence $M_k \to \infty$, let $a_{M_k} \to a$. For every fixed integer Δ, telescoping and boundedness of d_j give

$$\frac{1}{M_k} \sum_{j=0}^{M_k-1} \big(\phi(j+\Delta) - \phi(j)\big) \longrightarrow a\Delta,$$

with the finitely many boundary terms negligible.

Average (B) first over $j = 0, \dots, M_k - 1$, then over the product law $P \times Q$, and pass to the subsequence. Since

$$\mathbb{E}[I_Q - I_P] = H(Q) - H(P) = 0,$$

we obtain

$$c + \sum_x P(x)\lambda_x + \sum_w Q(w)\mu_w \le a\,\mathbb{E}[I_Q - I_P] = 0.$$

The gauge terms vanish, so $c \le 0$. □

PROVED. Any sharp lower certificate must depend on history, scale, time, or another nonlocal state variable. Merely replacing a periodic potential by an unbounded scalar potential cannot work.

23 Exact Hall formulation of one-step viability

The zero-repair recursion admits a useful prefix-capacity formulation. Let B have masses p_j and cumulative distribution

$$F(k) := \sum_{j \le k} p_j, \qquad F(k) = 0 \quad (k < 0).$$

After shifting the total information coordinate by four, define the source measure on $\{-1, 0, 1, \dots\}$ by

$$r_y := 3p_{y+1} + p_{y-1},$$

with cumulative mass

$$R(k) = 3F(k+1) + F(k-1).$$

A valid split is equivalent to choosing a low-branch submeasure $c \le r$ of total mass one. If $C(k) = \sum_{y \le k} c_y$, then

$$F_0(k+2) = C(k), \qquad F_1(k) = \frac{R(k) - C(k)}{3}. \tag{H}$$

Proposition 23.1 (Hall transfer criterion). *Let A^s, A^ℓ, A^h be nondecreasing CDF envelopes with limit one. Suppose a parent law satisfies $F \le A^s$ and*

$$3A^s_0 \le A^\ell_1, \tag{H0}$$

and, for every $k \ge 0$,

$$3A^s_{k+1} + A^s_{k-1} \le A^\ell_{k+2} + 3A^h_k. \tag{Hk}$$

Then the parent admits an exact zero-repair split whose low child satisfies $F_0 \le A^\ell$ and whose high child satisfies $F_1 \le A^h$.

Proof. Define

$$C(k) := \max\left\{R(-1),\ \max_{0 \le t \le k} \big(R(t) - 3A^h_t\big)\right\}.$$

Condition (H0) handles the forced boundary mass at $y = -1$. Condition (Hk), together with monotonicity of A^ℓ, gives $C(k) \le A^\ell_{k+2}$. By construction, $R(k) - C(k) \le 3A^h_k$. Moreover the running-maximum increments satisfy

$$0 \le C(k) - C(k-1) \le r_k,$$

because the increment of $R(k) - 3A^h_k$ is at most r_k. Thus C is the CDF of a submeasure $c \le r$. Since $R(k) - 3A^h_k \to 1$, we have $C(k) \to 1$. Equation (H) therefore defines probability laws B_0, B_1 and proves the claimed envelope bounds. □

Theorem 23.2 (No invariant single prefix-majorization envelope). *There is no CDF A on $\mathbb{Z}_{\ge 0}$ for which the cone*

$$\mathfrak{F}_A := \{F : F(k) \le A(k)\ \forall k\}$$

is closed under exact zero-repair splitting into two children that again belong to $\mathfrak{F}_A$.

Proof. Apply invariance to the probability law whose CDF is exactly A. Necessity of the child bounds gives

$$3A_0 \le A_1, \qquad 3A_{k+1} + A_{k-1} \le A_{k+2} + 3A_k.$$

Put $A_{-1} = 0$ and $d_k = A_k - A_{k-1}$. Then $d_k \ge 0$, $\sum_k d_k = 1$, and

$$d_{k+2} - 2d_{k+1} + d_k \ge 0.$$

Thus d is convex. A nonnegative summable convex sequence must be nonincreasing: if one discrete slope were positive, convexity would keep all later slopes bounded below by a positive number, contradicting $d_k \to 0$. Hence $d_1 \le d_0$. But $3A_0 \le A_1$ gives $d_1 \ge 2d_0$. Therefore $d_0 = d_1 = 0$, and nonincreasing nonnegativity forces $d_k = 0$ for all k, contradicting $\sum_k d_k = 1$. □

Theorem 23.3 (No finite family of finite-mean Hall envelopes). *Let S be finite. For each $s \in S$, let A^s be a CDF envelope with finite mean tail*

$$\sum_{k\ge 0} (1 - A^s_k) < \infty.$$

Assume there are maps $\ell, h : S \to S$ such that every type satisfies the Hall transfer inequalities

$$3A^s_0 \le A^{\ell(s)}_1,$$

$$3A^s_{k+1} + A^s_{k-1} \le A^{\ell(s)}_{k+2} + 3A^{h(s)}_k.$$

Then such a finite family cannot exist.

Proof. Restrict to a recurrent class of the finite transition chain that moves to $\ell(s)$ with probability $1/4$ and to $h(s)$ with probability $3/4$, and let π be its stationary law. Put

$$\beta^s_k := 1 - A^s_k, \qquad \beta^s_{-1} := 1,$$

and define the nonnegative Hall slacks

$$D_{s,k} := 3\beta^s_{k+1} + \beta^s_{k-1} - \beta^{\ell(s)}_{k+2} - 3\beta^{h(s)}_k,$$

$$E_s := 3\beta^s_0 - \beta^{\ell(s)}_1 - 2.$$

Finite means justify summing in k. With

$$\bar{\beta}_k := \sum_s \pi_s \beta^s_k, \qquad L_k := \sum_s \pi_s \beta^{\ell(s)}_k,$$

stationarity gives $L_k + 3H_k = 4\bar{\beta}_k$. A direct telescoping calculation yields

$$\sum_s \pi_s \sum_{k\geq 0} D_{s,k} = 1 - 3\bar{\beta}_0 + L_0 + L_1 = (L_0 - 1) - \sum_s \pi_s E_s.$$

The left side is nonnegative, whereas the right side is nonpositive because $L_0 \leq 1$ and $E_s \geq 0$. Hence equality holds everywhere. In particular $L_0 = 1$, every $E_s = 0$, and every $D_{s,k} = 0$ on the recurrent class. Consequently the envelope increments themselves satisfy the exact zero-repair recursion and form a finite exact mode automaton. This contradicts the complete finite-mode obstruction proved above. $\square$

PROVED. A successful Hall proof of H_1 must use a genuinely infinite family of envelopes. Neither one fixed CDF barrier nor finitely many finite-mean barriers can close.

24 Exact transform martingales and the cubic defect

For a node h, let $B_h(z)$ denote its PGF. Equation (Z) is equivalent to

$$\boxed{B_{h0}(z) + 3z^2 B_{h1}(z) = z(3 + z^2) B_h(z).} \tag{PGF}$$

For $0 < r < 1$, define

$$\alpha_0(r) := \frac{4}{r(3 + r^2)}, \qquad \alpha_1(r) := \frac{4r}{3 + r^2}.$$

If $H_n = X_1 \cdots X_n$ is sampled according to the P-history law $P(0) = 1/4$, $P(1) = 3/4$, then

$$\boxed{M_n(r) := B_{H_n}(r) \prod_{i=1}^{n} \alpha_{X_i}(r)}$$

is a nonnegative martingale. This follows immediately from (PGF).

A second exact form is obtained by exponentially tilting the output law:

$$\widetilde{P}_r(x) := \frac{P(x) r^{I_P(x)}}{\mu_P(r)},$$

where

$$\mu_P(r) = \frac{r^2(1 + 3r^2)}{4}, \qquad \mu_Q(r) = \frac{r^3(3 + r^2)}{4}.$$

Iterating (PGF) gives

$$\boxed{\mathbb{E}_{\widetilde{P}_r^n} B_{H_n}(r) = B_\varnothing(r) \left(\frac{\mu_Q(r)}{\mu_P(r)}\right)^n.} \tag{T}$$

The spectral gap is not asymptotic but exact:

$$\boxed{\mu_P(r) - \mu_Q(r) = \frac{r^2}{4}(1 - r)^3.} \tag{CubicGap}$$

Thus the critical transform scale is exactly $1 - r \asymp n^{-1/3}$.

24.1 Exact third-moment flux

Let

$$h = \frac{7}{2}, \qquad \xi := I_Q - h, \qquad \eta := I_P - h.$$

Then

$$\mathbb{E}\xi = \mathbb{E}\eta = 0, \qquad \mathbb{E}\xi^2 = \mathbb{E}\eta^2 = \frac{3}{4},$$

but

$$\mathbb{E}\xi^3 = \frac{3}{4}, \qquad \mathbb{E}\eta^3 = -\frac{3}{4}.$$

Hence

$$\mathbb{E}\xi^3 - \mathbb{E}\eta^3 = \frac{3}{2}.$$

For a history h, write

$$Y_h := \sum_{i \le |h|} (I_P(X_i) - 7/2),$$

and let

$$m_h = \mathbb{E}J_h, \qquad s_h = \mathbb{E}J_h^2, \qquad t_h = \mathbb{E}J_h^3.$$

Whenever the displayed moments are finite, the zero-repair identity implies that

$$m_{H_n}$$

is a martingale and so is

$$\boxed{s_{H_n} + 2Y_{H_n} m_{H_n}.}$$

At third order define

$$\Phi_h := t_h + 3Y_h s_h + 3Y_h^2 m_h = \mathbb{E}[(J_h + Y_h)^3] - Y_h^3.$$

Then the one-step drift is exactly

$$\boxed{\mathbb{E}[\Phi_{H_{n+1}} \mid H_n = h] - \Phi_h = \frac{3}{2} + \frac{9}{4} m_h.} \tag{Flux}$$

Consequently

$$\mathbb{E}\Phi_{H_n} = \Phi_\varnothing + n\left(\frac{3}{2} + \frac{9}{4} m_\varnothing\right)$$

whenever the third moments remain integrable.

Remark 24.1. *Equation* (Flux) *explains why a finite-mean H_1 construction, if it exists, must export the third-order mismatch into increasingly remote tails. The identity alone does not rule out finite mean, because the second and third moments may be infinite.*

ACTIVE RESEARCH FRONTIER. The sharp H_2 route is to truncate (Flux) into a family of linearly growing multiscale test functions whose expectations require only $\mathbb{E}J < \infty$. If the boundary errors are summable across dyadic scales, this would rule out every finite-mean element of $\mathcal{K}_\infty$. No such complete truncation theorem is currently proved.

25 Further compression no-go theorems

Theorem 25.1 (No finite-mean depth-only solution)**.** *Assume every node at depth t has the same law B_t. Then no infinite zero-repair solution can have $\mathbb{E}B_0 < \infty$.*

Proof. The recursion reduces to

$$\mu_Q * B_t = \mu_P * B_{t+1},$$

so for $0 < r < 1$,

$$B_{t+1}(r) = R(r)B_t(r), \qquad R(r) := \frac{r(3+r^2)}{1+3r^2} < 1,$$

where the strict inequality follows from (CubicGap). Hence $B_t(r) \to 0$. Therefore the buffer escapes to infinity in probability. On the other hand differentiating at $r = 1$ and using equal entropy gives

$$\mathbb{E}J_{t+1} = \mathbb{E}J_t.$$

A sequence of nonnegative laws escaping to infinity in probability cannot have a common finite mean. Contradiction. □

Theorem 25.2 (No score-only infinite solution)**.** *Suppose the buffer law depends only on the history score*

$$s(h) := 3N_0(h) - N_1(h).$$

Then no infinite zero-repair tree exists under this ansatz.

Proof. Write the corresponding PGFs as $B_s(z)$, $s \in \mathbb{Z}$. Since every integer score occurs at some depth, the recursion requires

$$B_{s+3}(z) + 3z^2 B_{s-1}(z) = z(3+z^2)B_s(z), \qquad s \in \mathbb{Z}. \tag{S}$$

Fix $r = 1/5$ and set $b_s = B_s(r)$. Then $0 < b_s \le 1$, while b_s satisfies a constant-coefficient recurrence with characteristic polynomial

$$p_r(y) = y^4 - r(3+r^2)y + 3r^2.$$

If $|y| \ge 1$ were a root, then

$$|y|^4 \le r(3+r^2)|y| + 3r^2 \le \big(r(3+r^2) + 3r^2\big)|y|.$$

For $r = 1/5$ the coefficient in parentheses is strictly below one, forcing $|y|^3 < 1$, a contradiction. Hence all characteristic roots satisfy $|y| < 1$. Every nonzero bi-infinite solution of the recurrence is therefore unbounded as $s \to -\infty$. This contradicts $0 < b_s \le 1$. □

PROVED. The ordering of the history is essential. Depth, output counts, or the single score $3N_0 - N_1$ do not retain enough information.

26 Boundary repair algebra

Away from the boundary $J = 0$, exact zero repair is elementary. If $B(0) = 0$, for example,

$$B \longmapsto (z^3 B, z^{-1}B)$$

is an exact split because

$$z^3 B + 3z^2(z^{-1}B) = z(3+z^2)B.$$

Thus all nontrivial geometry is generated when mass reaches the origin.

A useful local boundary correction is also exact.

Lemma 26.1 (One-step boundary repair). *Let*

$$B(z) = \sum_{j\ge 0} b_j z^j$$

be a PGF satisfying $b_2 \ge 2b_0$. *Then*

$$B_0(z) = zB(z) + 2b_0(z - z^3)$$

and

$$B_1(z) = \frac{2}{3}z^{-1}B(z) + \frac{1}{3}zB(z) - \frac{2b_0}{3}(z^{-1} - z)$$

are probability PGFs and form an exact zero-repair split.

Proof. The only potentially negative coefficient of B_0 is the coefficient of z^3, equal to $b_2 - 2b_0 \ge 0$; all other coefficients are nonnegative and $B_0(1) = 1$. In B_1 the apparent z^{-1} term cancels exactly, leaving

$$B_1(z) = \frac{2}{3}z^{-1}(B(z) - b_0) + \frac{1}{3}zB(z) + \frac{2b_0}{3}z,$$

which is manifestly nonnegative and normalized. Direct substitution gives (PGF). □

The mean-2 root

$$B_*(z) = \left(\frac{1+z}{2}\right)^4$$

satisfies $b_2 - 2b_0 = 1/4$, so this repair works at the root. However its high child has

$$b_0 = \frac{1}{6}, \qquad b_2 = \frac{1}{4},$$

and therefore violates $b_2 \ge 2b_0$. A single local repair rule cannot be iterated globally.

26.1 Geometric reservoir and the surviving hard spine

For

$$G_r(z) = \frac{1-r}{1-rz}, \qquad r \ge \sqrt{\frac{2}{3}},$$

the polynomially modified profile

$$F_r(z) := (3z - 2z^3)G_r(z)$$

is a PGF, and

$$\boxed{z(3+z^2)G_r = F_r + 3z^2(zG_r).}$$

Hence

$$G_r \longrightarrow (F_r, zG_r).$$

The high child then admits the bulk split

$$zG_r \longrightarrow (z^4G_r, G_r),$$

so the all-high branch returns to G_r after two high outputs. The true obstruction is therefore the low side-profile F_r, not the all-high spine. Finite collections of polynomial shapes do not absorb F_r uniformly as $r \uparrow 1$.

ACTIVE RESEARCH FRONTIER. The leading H_1 programme is now a scale-switching boundary repair: when a local condition such as $b_2 \ge 2b_0$ fails, the defect must be moved to the next dyadic scale rather than repaired within one fixed shape. The Harvey–Peres two-block residual has mass $1/4$ and doubles the information scale; more general atomic smoothing gives residual mass m^{-3} at scale m. The resulting mean budget is summable, but a fully one-symbol, prefix-compatible positive routing has not yet been constructed.

27 The mean-two affine lift is ruled out

The finite-horizon data exhibited the exact coincidence

$$\frac{2}{3}\mathcal{B}_{10} + \frac{2}{3} = \frac{2}{3}\frac{1346}{729} + \frac{2}{3} = \frac{4150}{2187} = \mathcal{B}_{11}.$$

This motivated the former candidate inequality

$$\mathcal{B}_{n+1} \le \frac{2}{3}\mathcal{B}_n + \frac{2}{3}. \tag{Affine}$$

The new infinite-viability lower bound kills this candidate globally.

Corollary 27.1 (No universal mean-two affine lift)**.** *The inequality* (Affine) *cannot hold for all n.*

Proof. If (Affine) held for all n, iteration from the finite initial value would give

$$\mathcal{B}_n \le 2 \qquad \forall n.$$

Hence $\sup_n \mathcal{B}_n \le 2$. By the compactness lemma below this would produce an infinite viable zero-repair tree with root mean at most 2. The exact 19/9 lower certificate proves that every such infinite tree has root mean at least $19/9 > 2$, a contradiction. □

The equality at the step $10 \to 11$ is therefore a finite-horizon coincidence, not the beginning of a global contraction law.

28 The viability-kernel formulation

The history labels can be eliminated conceptually.

Definition 28.1 (Finite-depth viability sets)**.** *Let*

$$\mathcal{K}_0 = \mathcal{P}(\mathbb{Z}_{\ge 0}),$$

the set of all probability laws on nonnegative integers.

Recursively, define

$$B \in \mathcal{K}_{n+1}$$

iff there exist

$$B_0, B_1 \in \mathcal{K}_n$$

such that

$$\boxed{\mu_Q * B = \frac{1}{4}\delta_2 * B_0 + \frac{3}{4}\delta_4 * B_1.}$$

Then

$$\mathcal{K}_{n+1} \subseteq \mathcal{K}_n,$$

and

$$\boxed{\mathcal{B}_n = \inf_{B \in \mathcal{K}_n} \mathbb{E}_B J.}$$

Define the infinite viability kernel

$$\boxed{\mathcal{K}_\infty := \bigcap_{n \ge 0} \mathcal{K}_n.}$$

The dichotomy H_1/H_2 becomes:

$$\boxed{H_1 : \quad \exists B \in \mathcal{K}_\infty \text{with } \mathbb{E}_B J < \infty,}$$

versus

$$\boxed{H_2 : \quad \mathcal{K}_\infty \text{contains no finite-mean law.}}$$

29 Compactness lemma

The viability formulation yields a useful existence reduction.

Lemma 29.1 (Bounded finite horizons imply an infinite tree). *Suppose*

$$\sup_n \mathcal{B}_n < \infty.$$

Then there exists an infinite zero-repair history-indexed tree

$$\{B_h\}_{h\in\{0,1\}^{<\infty}}$$

s finite root mean satisfying (Z) *at every node.*

Proof. Choose depth-n feasible trees with root means uniformly bounded by some constant C.

For every fixed history h, let

$$p_h = \left(\frac{1}{4}\right)^{N_0(h)} \left(\frac{3}{4}\right)^{N_1(h)} > 0$$

be its P-history probability.

The mean-buffer martingale identity implies

$$\sum_{|h|=t} p_h \, \mathbb{E} J_h = \mathbb{E} J_\varnothing \le C.$$

Hence for each fixed history

$$\mathbb{E} J_h \le \frac{C}{p_h}.$$

Therefore, for each fixed h, the family of laws appearing at node h over all sufficiently deep finite trees is tight by Markov's inequality. Since the binary tree is countable, a diagonal subsequence gives weak limits for every fixed node.

Convolution by the finite laws μ_P, μ_Q is continuous under weak convergence, and the linear identity (Z) is closed. Therefore the limits satisfy (Z) at every node. The root limit has mean at most C by lower semicontinuity. ☐

Corollary 29.2. *To prove H_1, it is enough to establish a uniform finite-horizon bound*

$$\boxed{\mathcal{B}_n \le C \quad \text{for all } n.}$$

To prove H_2, it is enough to show

$$\boxed{\mathcal{B}_n \to \infty.}$$

This remains the cleanest global decision problem, but the admissible finite branch is now quantitatively separated from 2.

30 Exact infinite-viability lower certificates: 17/9, 55/27, 19/9

Let

$$B(z) = \sum_{j\ge 0} b_j z^j$$

be the buffer law at an arbitrary node of an infinite viable zero-repair tree, and write

$$F_B(k) = \sum_{j=0}^{k} b_j.$$

Because every child of an infinite viable node is again infinite viable, low information tail inequalities may be propagated recursively through the zero-repair equations.

30.1 Closed CDF envelope and the $17/9$ bound

Lemma 30.1 (Invariant three-level CDF envelope). *Every node of an infinite viable zero-repair tree satisfies*

$$\boxed{F_B(0) \le \frac{1}{9}, \qquad F_B(1) \le \frac{1}{3}, \qquad F_B(2) \le \frac{2}{3}.} \tag{A}$$

Proof. Suppose all children satisfy temporary caps

$$F(0) \le q_0, \qquad F(1) \le q_1, \qquad F(2) \le q_2.$$

Writing the zero-repair coefficient equations at total-information levels $u = 3, 4, 5$ gives the backward implications

$$F_B(0) \le \frac{1}{3}q_1,$$

$$F_B(1) \le q_0 + \frac{1}{3}q_2,$$

and

$$F_B(2) \le q_1 + \frac{1}{3}.$$

Thus the cap vector is propagated by

$$T(q_0, q_1, q_2) = \left(\frac{1}{3}q_1, q_0 + \frac{1}{3}q_2, q_1 + \frac{1}{3}\right).$$

Starting from the trivial finite-depth cap $(1, 1, 1)$ and iterating backward through arbitrarily many viable generations is legitimate for an infinite tree. The affine map has the unique fixed point

$$q_* = \left(\frac{1}{9}, \frac{1}{3}, \frac{2}{3}\right),$$

and the linear part has eigenvalues $0, \pm\sqrt{2/3}$, so $T^m(1, 1, 1) \to q_*$. Passing to the limit yields (A). $\square$

Corollary 30.2 (First support-independent lower bound). *Every root of an infinite viable tree satisfies*

$$\boxed{\mathbb{E}J \ge \frac{17}{9}.}$$

Proof. Using the tail formula

$$\mathbb{E}J = \sum_{k \ge 0} \mathbb{P}(J > k)$$

and only the first three terms,

$$\mathbb{E}J \ge \sum_{k=0}^{2} (1 - F_B(k)) \ge \left(1 - \frac{1}{9}\right) + \left(1 - \frac{1}{3}\right) + \left(1 - \frac{2}{3}\right) = \frac{17}{9}.$$

$\square$

30.2 Combined child viability and the $55/27$ bound

Two further inequalities follow directly from one more viable generation. Define

$$C_1(B) := b_0 + \frac{3}{4}(b_1 + b_2), \qquad C_2(B) := b_0 + b_1 + \frac{3}{4}(b_2 + b_3).$$

Lemma 30.3 (One-generation combined-tail constraints)**.** *Every node of an infinite viable tree satisfies*

$$\boxed{C_1(B) \le \frac{1}{2}, \qquad C_2(B) \le \frac{3}{4}.} \tag{C}$$

Proof. Let L, H be the low and high children. The coefficient equations at $u = 3, 4, 5, 6$ are

$$3b_0 = l_1,$$

$$3b_1 = l_2 + 3h_0,$$

$$3b_2 + b_0 = l_3 + 3h_1,$$

$$3b_3 + b_1 = l_4 + 3h_2.$$

Summing the first three in the combination appropriate to C_1 gives

$$4C_1(B) = F_L(3) + 3F_H(1) \le 1 + 3 \cdot \frac{1}{3} = 2,$$

using (A) for the high child. Likewise summing all four gives

$$4C_2(B) = F_L(4) + 3F_H(2) \le 1 + 3 \cdot \frac{2}{3} = 3.$$

□

Proposition 30.4 (One-generation dual bound)**.** *Every node of an infinite viable tree satisfies*

$$\boxed{\mathbb{E}J \ge \frac{55}{27}.}$$

Proof. For $j \ge 0$ define

$$c_1(j) = \begin{cases} 1, & j = 0, \\ 3/4, & j = 1, 2, \\ 0, & j \ge 3, \end{cases} \qquad c_2(j) = \begin{cases} 1, & j = 0, 1, \\ 3/4, & j = 2, 3, \\ 0, & j \ge 4. \end{cases}$$

The pointwise inequality

$$j \ge 4 - \frac{2}{3}\mathbf{1}_{\{j \le 0\}} - \frac{2}{3}\mathbf{1}_{\{j \le 1\}} - \frac{4}{3}c_1(j) - \frac{4}{3}c_2(j)$$

is checked directly for $j = 0, 1, 2, 3$ and is trivial for $j \ge 4$. Taking expectations and applying (A) and (C) gives

$$\mathbb{E}J \ge 4 - \frac{2}{3}\frac{1}{9} - \frac{2}{3}\frac{1}{3} - \frac{4}{3}\frac{1}{2} - \frac{4}{3}\frac{3}{4} = \boxed{\frac{55}{27}}.$$

□

30.3 Exact two-generation dual certificate: 19/9

The preceding inequalities can be inserted simultaneously at a parent and its two children. This yields a support-independent exact rational dual.

Theorem 30.5 (Exact 19/9 infinite-viability lower bound)**.** *Let B be the root of any infinite zero-repair history-indexed tree for the $R = 2$ parity pair. Then*

$$\boxed{\mathbb{E}_B J \geq \frac{19}{9} > 2.} \tag{19/9}$$

Consequently

$$\boxed{\mathcal{B}_\infty \geq \frac{19}{9} > 2.}$$

Proof. Let L, H be the two children of B. Use the active viability inequalities

$$\begin{array}{lll} \text{parent:} & A_0(B) \leq 1/9, & A_1(B) \leq 1/3, \\ \text{low child:} & A_2(L) \leq 2/3, & C_2(L) \leq 3/4, \\ \text{high child:} & A_1(H) \leq 1/3, & C_1(H) \leq 1/2, \end{array}$$

where $A_k(B) = F_B(k)$, together with normalization and the exact zero-repair coefficient equalities.

Assign to the total-information equality at level u the multiplier

$$\lambda_u = \begin{cases} -20/9, & u = 3, \\ -16/9, & u = 4, \\ -4/3, & u = 5, \\ 0, & u = 6, \\ 4/3, & u \geq 7, \\ 0, & u < 3. \end{cases}$$

Use normalization multipliers $8/3$ for the parent and 1 for the high child, and inequality multipliers

$$-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{9}, -\frac{4}{9}, -1, -\frac{4}{3}$$

for, respectively,

$$A_0(B), A_1(B), A_2(L), C_2(L), A_1(H), C_1(H).$$

After collecting coefficients of the variables b_j, l_j, h_j, the resulting dual coefficient triple (p_j, ℓ_j, r_j) is

j	p_j	ℓ_j	r_j
0	0	$-5/9$	0
1	1	0	0
2	2	0	0
3	3	0	0
4	4	0	0
$j \geq 5$	4	$-1/3$	0.

Hence

$$p_j \leq j, \qquad \ell_j \leq 0, \qquad r_j \leq 0 \quad (j \geq 0).$$

This verifies dual feasibility on the entire infinite support; the tail is explicit and no truncation is used.

The dual objective is exactly

$$\frac{8}{3} + 1 - \frac{1}{3}\frac{1}{9} - \frac{1}{3}\frac{1}{3} - \frac{1}{9}\frac{2}{3} - \frac{4}{9}\frac{3}{4} - 1 \cdot \frac{1}{3} - \frac{4}{3}\frac{1}{2} = \boxed{\frac{19}{9}}.$$

Therefore weak duality gives (19/9).

Finally, if $\mathcal{B}_\infty < 19/9$, the compactness lemma would produce an infinite viable tree with root mean below 19/9, contradicting the bound just proved. Hence $\mathcal{B}_\infty \geq 19/9$. □

Corollary 30.6 (Mean two is impossible)**.** *There is no $B \in \mathcal{K}_\infty$ with $\mathbb{E}_B J = 2$.*

Corollary 30.7 (Finite-horizon values must eventually exceed two)**.** *Because $\mathcal{B}_n$ is nondecreasing and*

$$\lim_{n\to\infty} \mathcal{B}_n \geq \frac{19}{9},$$

there exists a finite n for which

$$\mathcal{B}_n > 2.$$

ACTIVE RESEARCH FRONTIER. The lower hierarchy is now genuinely active mathematics rather than speculation. The immediate dual question is whether the closed-tail construction can be iterated to produce a strictly increasing sequence of support-independent bounds beyond 19/9, and whether that sequence diverges or converges to a finite constant. The constructive H_1 question is now harder as well: any infinite repair must have root mean at least 19/9, so all mean-2 constructions are definitively excluded.

31 Why the old $n^{1/3}$ renormalization target is premature

The self-similar two-block residual suggests a formal scale relation: information is dilated by 2, residual mass is reduced by 1/4, and two-symbol blocking contributes another factor 2 in causal time.

This points to the candidate scale transformation

$$n \mapsto 8n, \qquad J \mapsto 2J.$$

If one could prove

$$\mathcal{B}_{8n} \gtrsim 2\mathcal{B}_n,$$

then $n^{1/3}$ would follow.

The exact lower bound $\mathcal{B}_\infty \geq 19/9$ rules out convergence to 2. More generally, if $\mathcal{B}_n$ converges to any finite positive limit $L \geq 19/9$, then a multiplicative inequality of the form $\mathcal{B}_{8n} \gtrsim 2\mathcal{B}_n$ is asymptotically impossible.

Therefore the correct order of attack is still:

1. decide whether $\mathcal{K}_\infty$ contains a finite-mean law;
2. only if the answer is negative, return to a quantitative renormalization lower bound and determine the divergence exponent.

32 Prefix-code interpretation

The $R = 2$ pair has a concrete binary-prefix representation.

For P, take one codeword of length 2 and twelve codewords of length 4:

$$2^{-2} + 12\,2^{-4} = 1.$$

For Q, take six codewords of length 3 and eight codewords of length 5:

$$6\,2^{-3} + 8\,2^{-5} = 1.$$

The length decisions may be coupled through the same first two fair bits:

$$00: \qquad L_P = 2, \quad L_Q = 5,$$

and

$$\neq 00: \qquad L_P = 4, \quad L_Q = 3.$$

Thus the third-order parity-ladder problem can be viewed as

$$\boxed{\text{causal synchronization of two complete prefix parsers}}$$

with identical average code length

$$\mathbb{E}L_P = \mathbb{E}L_Q = 3.5,$$

identical variance, but distinct third-order information structure.

This interpretation connects the zero-repair buffer directly to finitary-code constructions.

33 Relation to finitary coding

Harvey and Peres prove that, for equal-entropy Bernoulli shifts, finite square-root coding-length moment forces equality of informational variance. This is the coding-theoretic analogue of the sharp second-order causal obstruction.

Their ordered measure-preserving matching lemma also converts generating function self-similarity into a positive residual matching. Hence the remaining obstruction in the present programme is not static positivity but unilateral compatibility across future histories.

Gabor's 2024 work proves moment-optimal finitary isomorphism for equal-entropy aperiodic Markov processes, in particular iid sources: coding-radius moments are achievable for every

$$\theta < \frac{1}{2}$$

in full generality, up to polylogarithmic effects.

Gabor's 2025 work extends the Harvey–Peres variance obstruction to $\mathbb{Z}^d$ and proves a Schmidt-type rigidity statement: exponentially decaying coding-radius tails at equal entropy force equality of one-site information laws up to permutation. The same work obtains equality of all information moments under exponential-tail coding.

These results do not settle the present finite-mean zero-repair viability question, but they strongly suggest that any H_1 construction must be heavy-tailed and genuinely multiscale rather than finite-state or exponential-tail.

34 Relation to causal optimal transport

Discrete-time causal optimal transport admits dynamic programming and duality formulations. General adapted-transport dual attainment is also known.

Therefore the difficulty of the current lower problem is not the existence of a dual formulation. The difficulty is identifying an explicit multiscale dual or viability invariant adapted to the exact parity-ladder finite-difference self-similarity.

Recent continuous-time causal-transport work formulates analogous problems as state-constrained stochastic control and nonlinear master equations, reinforcing the interpretation of the buffer as an information inventory state.

35 Computational audit

All computational checks used in this attack are deterministic; no Monte Carlo evidence is used as proof.

The executable audits include:

- exact finite construction of sorted-arithmetic causal couplings;
- exhaustive action-path verification for small horizons;
- exact monotone block transport;
- dyadic finite-bit exactization audits;
- exact parity-ladder moment and generating-function identities;
- global zero-repair history-tree LPs;
- finite-support stability checks;
- finite-state and finite-memory crash tests;
- exact atomic B-spline residual enumeration;
- Fourier stationary-buffer lower certificates.

The role of computation is to expose and falsify finite claims. The asymptotic statements are accepted only when supported by analytic proof.

36 Current theorem ledger

Statement	**Status**
Projective infinite causal substrate at rate C_∞^{causal}	proved
Anti-diagonal causal stream-to-tree lift	proved
$E_n = \sqrt{2/\pi}\lvert\sigma_P - \sigma_Q\rvert\sqrt{n} + o(\sqrt{n})$	proved
Parity-ladder first-unmatched-moment construction	proved
$\mu_{P_R} - \mu_{Q_R} = 2^{-(r-1)}\Delta^r$	proved
Fully atomic B-spline $O(n^{1/r})$ causal upper bound	proved
Stationary independent-buffer $\Omega(n^{1/r})$	proved
Static positive residual extraction	proved
No stationary integrable exact zero-repair state	proved
No finite exact zero-repair automaton for any finite K	proved
No positive scalar state-only Bellman drift	proved
Exact Hall transfer criterion	proved
No invariant single prefix-majorization Hall envelope	proved
No finite family of finite-mean Hall envelopes	proved
Exact transform martingale and cubic transform gap	proved
Exact third-moment flux law, under finite third moments	proved
No finite-mean depth-only zero-repair solution	proved
No score-only infinite zero-repair solution	proved
One-step boundary-repair lemma	proved
Invariant CDF caps $(1/9, 1/3, 2/3)$	proved
Infinite-tree lower bound $\mathbb{E}J \geq 17/9$	proved
Combined-tail lower bound $\mathbb{E}J \geq 55/27$	proved
Exact support-independent lower bound $\mathbb{E}J \geq 19/9$	**proved**
$\mathcal{B}_\infty \geq 19/9 > 2$	**proved**
Existence of $B \in \mathcal{K}_\infty$ with $\mathbb{E}J = 2$	**ruled out**

Statement	Status
Universal affine lift $\mathcal{B}_{n+1} \le \frac{2}{3}\mathcal{B}_n + \frac{2}{3}$	**ruled out**
Zero-repair LP data through depth 11	audited evidence
Mean-2 roots feasible through depth 11	finite-depth evidence only
$19/9 \le \mathcal{B}_\infty < \infty$	open
$\mathcal{B}_n \to \infty$	open
$\mathcal{B}_n = \Theta(n^{1/3})$	not established
$E_n = \Theta(n^{1/r})$ for the full C3-C parity ladder	open

37 The present decision tree

The programme remains bifurcated, but the finite branch has moved strictly above two.

Branch H_1: bounded viability above the exact barrier

Prove

$$\sup_n \mathcal{B}_n < \infty.$$

By compactness this yields an infinite history-indexed zero-repair tree with finite root mean. The new theorem imposes the non-negotiable constraint

$$\boxed{\frac{19}{9} \le \mathcal{B}_\infty < \infty.}$$

Any constructive invariant family must therefore live strictly above the former mean-2 regime.

ACTIVE RESEARCH FRONTIER. This is the main constructive attack. Find an order-sensitive, infinitely multiscale boundary-repair hierarchy whose root mean is finite and at least 19/9. All finite-mode closures, all finite Hall-envelope families, and all mean-2 constructions are now analytically unavailable.

Branch H_2: divergent viability

Prove

$$\mathcal{B}_n \to \infty.$$

The exact 19/9 dual shows that low-tail viability constraints already create a genuine positive asymptotic barrier. The immediate question is whether further closed-tail generations continue to raise that barrier without bound. The cubic flux and transform identity remain complementary multiscale tools, but they are no longer the only known lower mechanism.

ACTIVE RESEARCH FRONTIER. This is now the main lower-bound attack: iterate the exact closed-tail/LP-dual hierarchy beyond two generations. Either the sequence of support-independent certificates diverges, proving H_2, or it saturates at a finite value and supplies a quantitative target for an H_1 construction.

38 Immediate research targets

R1. Push the exact dual beyond 19/9. Derive the next closed-tail viability inequalities and solve the exact rational dual at three, four, and more generations. Determine whether the certified lower sequence is monotone and whether it diverges.

R2. Search for finite viability above $19/9$**.** Construct a countable scale-indexed family of buffer laws with a finite root mean $C \geq 19/9$ and verify the zero-repair recursion at every history.

R3. Track the first crossing above two. The exact theorem implies that some finite-horizon optimum satisfies $\mathcal{B}_n > 2$. Computing the first such n would identify where the pre-asymptotic mean-2 mirage breaks and may expose the next active low-tail constraints.

R4. Integrate the closed-tail dual with the cubic multiscale flux. The rational low-level certificate and the transform/cubic identities probe different parts of the state space. A combined dual may be stronger than either mechanism alone.

R5. Only after H_1/H_2 **is settled, return to the full entropy bill.** A quantitative comparison is then required between positive repair entropy and the zero-repair viability functional.

39 Bottom line

The sharp second-order problem is closed. The higher-order parity ladder has a rigorous $O(n^{1/r})$ causal upper hierarchy and a matching lower theorem inside the stationary independent-buffer architecture. For the full history-indexed $r = 3$ zero-repair problem, the structural attack has ruled out stationary integrable recycling, greedy continuation, finite automata, scalar Bellman drift, depth-only and score-only compression, one fixed Hall barrier, and every finite family of finite-mean Hall barriers.

The latest lower-bound attack changes the quantitative frontier. Infinite viability forces the exact low-tail constraints

$$F(0) \leq \frac{1}{9}, \qquad F(1) \leq \frac{1}{3}, \qquad F(2) \leq \frac{2}{3},$$

which yield

$$\mathbb{E}J \geq \frac{17}{9}.$$

One more generation gives

$$\mathbb{E}J \geq \frac{55}{27},$$

and the exact two-generation support-independent dual gives

$$\boxed{\mathbb{E}J \geq \frac{19}{9} > 2.}$$

Therefore

$$\boxed{\mathcal{B}_\infty \geq \frac{19}{9} > 2,}$$

and the former mean-2 hypothesis is dead.

The decisive question is now

$$\boxed{\frac{19}{9} \leq \mathcal{B}_\infty < \infty \quad \text{or} \quad \mathcal{B}_n \to \infty\ ?}$$

This is the current research frontier. The next mathematically meaningful move is not another mean-2 ansatz; it is either a higher-generation exact dual certificate or an explicit infinite repair construction above the $19/9$ barrier.

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